

SwiftBot ZigZag Navigation System

Software Requirements Specification (SRS)

Author: Muhammed Oluwafemi Olafusi

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Introduction

This document defines the functional and non-functional requirements for the SwiftBot ZigZag Navigation System.

The system enables a robot to perform a zigzag traversal based on QR code input, validate parameters, execute movement, and return to its starting position while logging performance data.

The purpose of this document is to provide a clear specification for system design and implementation.

Functional Requirements

1. When the system starts, the system should prompt the user to scan a QR code.
2. The system should accept two values encoded in the format Value1-Value2.
3. The system should interpret Value1 as the zigzag section length (cm).
4. The system should interpret Value2 as the number of zigzag sections.
5. The system should validate that Value1 is between 15 cm and 85 cm inclusive.
6. The system should validate that Value2 is even and does not exceed 12.
7. If the QR input is invalid, the system should display an error message and reject the input.
8. Once validated, the system should display the two user inputs.
9. The system should generate a random wheel speed within the safe operating range.
10. The system should calculate the movement duration for one zigzag section using the wheel speed and the user-provided length.
11. The system should set the LED to green during the first section.
12. The system should move forward for one calibrated section.
13. The system should stop for one second before turning.
14. After each odd-numbered section, the system should turn 90 degrees to the left.

15. After each even-numbered section, the system should turn 90 degrees to the right.
16. The LED should alternate between green and blue for each successive section.
17. Steps 12–16 should repeat until all zigzag sections are completed.
18. After completing the zigzag, the system should turn 180 degrees to face the start position.
19. The system should retrace each zigzag section in reverse order.
20. The system should display LED colours matching the corresponding forward sections.
21. The system should continue retracing until it reaches the original starting position.
22. The system should stop and turn off the LEDs at the end of the return journey.
23. The system should write the following information to a log file:
 - 23.1. zigzag section length,
 - 23.2. number of sections,
 - 23.3. random wheel speed,
 - 23.4. total distance travelled (forward only),
 - 23.5. total duration of the forward journey,
 - 23.6. straight-line distance between start and end.
24. The system should store the log file in an accessible location.
25. After logging, the system should prompt the user to scan another QR code or terminate.
26. If the user enters 'Y', the system should restart the process.
27. If the user enters 'X', the system should begin termination.
28. The system should maintain a count of completed zigzag journeys.
29. The system should track the journey with the longest straight-line distance.
30. The system should track the journey with the shortest straight-line distance.
31. When the user chooses to exit, the system should display:
 - 31.1. number of journeys completed,
 - 31.2. the longest journey,
 - 31.3. the shortest journey,

31.4. the log file location.

32. After displaying the summary, the system should terminate.

Additional Functionality

33. After validating the QR input, the system should prompt the user to choose a speed profile.

34. The system should offer three speed profiles: Slow, Normal, Fast.

35. The system should accept 'S', 'N', or 'F' as valid input options.

36. Invalid choices should cause an error message and re-prompt.

37. The Slow profile should generate wheel speeds in the lower safe range.

38. The Normal profile should generate wheel speeds in the middle range.

39. The Fast profile should generate wheel speeds in the upper safe range.

40. The selected profile should determine the wheel speed used for all timing calculations.

41. The system should display the final chosen speed and profile before movement begins.

42. The log file should include the selected speed profile.

43. If no profile is selected, the system should default to Normal.

Non-Functional Requirements

44. The system should process QR input within one second.

45. Error messages should be clear and informative.

46. LED colours should remain visible from at least one metre.

47. Movement should remain stable and safe during all sections.

48. The system should not crash during standard operation.

49. The design should allow future extension or modification.

This additional functionality improves usability by allowing users to adapt the robot's behaviour to different environments while preserving all required core functionality